



## PROBABILITY AND STATISTICS

### QUESTION BANK

#### MODULE 1

#### SHORT QUESTION AND ANSWER

Q1. Calculate mean from the following data

|   |   |    |    |    |
|---|---|----|----|----|
| x | 5 | 10 | 15 | 20 |
| f | 4 | 6  | 8  | 2  |

SOLUTION:

| x            | f             | xf              |
|--------------|---------------|-----------------|
| 5            | 4             | 20              |
| 10           | 6             | 60              |
| 15           | 8             | 120             |
| 20           | 2             | 40              |
| <b>Total</b> | $\sum f = 20$ | $\sum fx = 240$ |

$$\bar{X} = \frac{\sum fx}{\sum f} = \frac{240}{20} = 12, \sum f = \text{sum of all the frequencies}$$



## Q2. Define mean

### Mean

Mean is defined as average of the data. Mean is further divided into three kinds, which are the arithmetic mean, the geometric mean and the harmonic mean.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{N} = \frac{\Sigma X}{N}$$

Where,  $\bar{X}$  = Arithmetic Mean;  $\Sigma x$  = Sum of all the values of the variables i.e.,  $X_1 + X_2 + X_3 + \dots + X_n$   $N$  = Number of observations.

## Q3. Write a short note on Geometric mean.

### **Geometric mean :**

The geometric mean of a series containing  $n$  observations is the  $n^{\text{th}}$  root of the product of the values. If  $x_1, x_2, \dots, x_n$  are observations then

$$\begin{aligned} \text{G.M} &= \sqrt[n]{x_1 \cdot x_2 \dots x_n} \\ &= (x_1 \cdot x_2 \dots x_n)^{1/n} \\ \log \text{GM} &= \frac{1}{n} \log(x_1 \cdot x_2 \dots x_n) \\ &= \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n) \\ &= \frac{\Sigma \log x_i}{n} \\ \text{GM} &= \text{Antilog} \frac{\Sigma \log x_i}{n} \end{aligned}$$

For grouped data

$$\text{GM} = \text{Antilog} \left[ \frac{\Sigma f \log x_i}{N} \right]$$

## Q4. Write a short note on Harmonic mean.



**Q5. Define range. Find the range of weights of 7 students from the following. 27, 30, 35, 36, 38, 40, 43**

Range

Range is the simplest and crudest measure of dispersion. It is a rough measure of dispersion. It is the difference between the highest and the lowest

**Harmonic mean (H.M) :**

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic average of the reciprocal of the given values. If  $x_1, x_2, \dots, x_n$  are  $n$  observations,

$$H.M = \frac{n}{\sum_{i=1}^n \left( \frac{1}{x_i} \right)}$$

For a frequency distribution

$$HM = \frac{N}{\sum_{i=1}^n f \left( \frac{1}{x_i} \right)}$$

value in the distribution.

**Range = L – S**

Where, L = Largest Value; S = Smallest Value.

The Relative measure of range is called as the Co – efficient of Range.

**Co – efficient of Range =  $\frac{L-S}{L+S}$**

Range = L – S Here L = 43; S = 27

$\therefore$  Range = 43 – 27 = 16

**Co – efficient of Range =  $\frac{L-S}{L+S}$  =  $\frac{43-27}{43+27}$  = 16 = 0.23**



**Q6. Find out the median of the following items. X: 10, 15, 9, 25, 19.**

**SOLUTION:**

| S. No. | Size of ascending order |
|--------|-------------------------|
| 1      | 9                       |
| 2      | 10                      |
| 3      | 15                      |
| 4      | 19                      |
| 5      | 25                      |

$$\begin{aligned}\text{Median} &= \text{Size of } \frac{(N+1)}{2} \text{th term} &= \text{Size of } \frac{(5+1)}{2} \text{th term} \\ & &= 3^{\text{rd}} \text{ item} = 15\end{aligned}$$

**Q7. Define median, mode.**

### Median

Median is the single value that measures the central item in the data. It may be defined as the value of that item which divides the series into equal parts, one half containing values greater than it and the other half containing values less than it.

### Mode

Mode is the value occurring more frequency in a set of observations

**Q8. Find the IQR for the data: 2, 4, 6, 8, 10, 12, 14, 16, 18**

**Solution: (n = 9)**

$$\begin{aligned}\left[ Q_1 &= \frac{n+1}{4} = \frac{10}{4} = 2.5^{\text{th}} \text{ item} \right] \\ \left[ Q_3 &= \frac{4+6}{2} = 5 \right]\end{aligned}$$



$$\left[ Q_3 = \frac{3(n+1)}{4} = \frac{3(10)}{4} = 7.5^{th} \text{ item} \right]$$

$$\left[ Q_3 = \frac{14 + 16}{2} = 15 \right]$$

$$[IQR = Q_3 - Q_1 = 15 - 5 = 10]$$

$$IQR = 10$$

## Long question and answers

### Q1 . What is Statistics, Characteristics of Statistics ,merits and demerits.

- Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data. In other words, it is a mathematical discipline to collect, summarize data.
- According to Croxton and Cowden, Statistics may be defined as the science of collection, presentation, analysis and interpretation of numerical data.

#### Nature of statistics

- Statistics is both a science and an art
- As a science statistical methods are generally systematic and based on fundamental ideas and processes
- It also works as a base for all other sciences.

#### Scope of statistics

- Statistics is a mathematical science pertaining to the collection, analysis, interpretation or explanation and presentation of data.
- It provides tools for predicting and forecasting the economic activities.
- It is useful for an academician, government, business etc.

#### Characteristics of statistics

- Statistics are Aggregate of facts
- It is numerically expressed



- The statistical Data affected by multiplicity of causes
- It is enumerated according to reasonable standard of accuracy
- It is collected in systematic accuracy
- It is collected for pre-determined purpose
- It is placed in relation to other

### **Merits of statistics**

- Statistics helps in
- Providing a better understanding
- Exact description
- Collecting appropriate quantitative data
- Business forecasting
- Decision making
- Quality control

### **Limitations of statistics**

- It is not useful for individual cases
- It ignores qualitative aspects
- It deals with average only
- Improper use of statistics can be dangerous

### **Q2.From the following find out the mean using assumed mean method:**

| Class Interval | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 - 50 |
|----------------|--------|---------|---------|---------|---------|
| Frequency      | 6      | 5       | 8       | 15      | 7       |

### **SOLUTION:**

For a **Continuous Series**, the **Assumed Mean Method** formula is  $\bar{X} = A + \frac{\sum fd}{N}$

Where,  $\bar{X}$  = Arithmetic Mean; A = Assumed mean;  $\sum fd$  = Sum of total deviations; N = Total frequency.

| Class Interval<br>$x_i$ | Mid Point<br>( $x_i$ ) | $d = x_i - A$ | F | fd |
|-------------------------|------------------------|---------------|---|----|
|-------------------------|------------------------|---------------|---|----|



|         |    |                |                                       |                                      |
|---------|----|----------------|---------------------------------------|--------------------------------------|
| 0 – 10  | 5  | 5 – 25 = -20   | 6                                     | -120                                 |
| 10 – 20 | 15 | 15 – 25 = - 10 | 5                                     | -50                                  |
| 20 – 30 | 25 | 25 – 25 = 0    | 8                                     | 0                                    |
| 30 – 40 | 35 | 35 – 25 = 10   | 15                                    | 150                                  |
| 40 - 50 | 45 | 45 – 25 = 20   | 7                                     | 140                                  |
|         |    |                | <b><math>\Sigma f = N = 41</math></b> | <b><math>\Sigma fd = +120</math></b> |

$d = x_i - A$ ; here  $A = 25$

$$\bar{X} = A + \frac{\Sigma fd}{N}$$

$$= 25 + \frac{+120}{41}$$

$$25 + 2.93 = 27.93$$

**Q3.Consider the following distribution of daily wages of 50 workers of a factory**

|                           |                |                |                |                |                |
|---------------------------|----------------|----------------|----------------|----------------|----------------|
| <b>Daily wages (in ₹)</b> | <b>500-520</b> | <b>520-540</b> | <b>540-560</b> | <b>560-580</b> | <b>580-600</b> |
| <b>Number of workers</b>  | 12             | 14             | 8              | 6              | 10             |

**Find the mean daily wages of the workers of the factory by using an appropriate method.**



SOLUTION:

In this case, we can use **step-deviation method** because the value of  $x_i$  and  $f_i$  are large.

Class mark ( $x_i$ ) = (lower class limit + upper class limit) / 2

So let us assume the mean value,  $a = 550$  and **class interval**,  $h = 20$ .

So,  $d' = (x_i - a) / h = (x_i - 550) / 20$  where  $d'$  is the deviation taken from the observation.

| Daily wages<br>(Class interval) | N0. of workers<br>frequency ( $f_i$ ) | Class mark<br>( $x_i$ ) | $d' = (x_i - 550) / 20$ | $f d'$                                  |
|---------------------------------|---------------------------------------|-------------------------|-------------------------|-----------------------------------------|
| 500-520                         | 12                                    | 510                     | -2                      | -24                                     |
| 520-540                         | 14                                    | 530                     | -1                      | -14                                     |
| 540-560                         | 8                                     | 550 = a                 | 0                       | 0                                       |
| 560-580                         | 6                                     | 570                     | 1                       | 6                                       |
| 580-600                         | 10                                    | 590                     | 2                       | 20                                      |
| <b>Total</b>                    | <b><math>\Sigma f_i = 50</math></b>   |                         |                         | <b><math>\Sigma f_i d' = -12</math></b> |

the mean  $\bar{x}$  of the given data is given by

$$\bar{x} = a + (h \times \Sigma f d') / \Sigma f_i$$

$$\bar{x} = 550 + (20 \times (-12 / 50))$$

$$\bar{x} = 550 - 4.8 = 545.20$$



**Q4. From the following find out the mode**

| Class Interval | 10-25 | 25-40 | 40-55 | 55-70 | 70-85 | 85-100 |
|----------------|-------|-------|-------|-------|-------|--------|
| Frequency      | 6     | 20    | 44    | 26    | 3     | 1      |

**SOLUTION:**

$$\text{Mode} = l + \left[ \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \right] \times c$$

where  $f_1$  is the frequency of the modal class

$f_0$  is the frequency of the class preceding the modal class

$f_2$  is the frequency of the class succeeding the modal class  $c$  is the class interval of the modal class

$l$  is the lower limit of the modal class Modal class the class corresponding to the highest frequency.

| marks  | No. of students    |
|--------|--------------------|
| 10-25  | 6                  |
| 25-40  | 20 $f_0$           |
| 40-55  | 44 $f_1$           |
| 55-70  | 26 $f_2$           |
| 70-85  | 3                  |
| 85-100 | 1                  |
|        | $\Sigma f_i = 100$ |

Modal class is 40 -55

$$L = 40, f_1 = 44, f_0 = 20, f_2 = 26, c = 55 - 40 = 15.$$

$$\text{Mode} = l + \left[ \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \right] \times c$$

$$= 40 + \left[ \frac{(44 - 20)}{(2 \times 44 - 20 - 26)} \right] \times 15$$

$$= 40 + \left[ \frac{24}{(88 - 46)} \right] \times 15$$



$$= 40 + [(24/42) \times 15]$$

$$= 40 + [0.5714 \times 15]$$

$$= 40 + 8.57$$

$$= 48.57.$$

**Q5. Find out the geometric mean of the following data.**

| Yield of wheat (mounds) | No. of farms |
|-------------------------|--------------|
| 7.5 – 10.5              | 5            |
| 10.5 – 13.5             | 9            |
| 13.5 – 16.5             | 19           |
| 16.5 – 19.5             | 23           |
| 19.5 – 22.5             | 7            |
| 22.5 – 25.5             | 4            |
| 25.5 – 28.5             | 1            |

**SOLUTION:**

$$\text{G.M.} = \text{antilog of } \frac{\sum F \log m}{N}$$

| Yield of wheat (mounds) | No. of farms (f) | m  | log m  | f log m |
|-------------------------|------------------|----|--------|---------|
| 7.5 – 10.5              | 5                | 9  | 0.9542 | 4.7710  |
| 10.5 – 13.5             | 9                | 12 | 1.0792 | 9.7128  |
| 13.5 – 16.5             | 19               | 15 | 1.1761 | 22.3459 |
| 16.5 – 19.5             | 23               | 18 | 1.2553 | 28.8719 |
| 19.5 – 22.5             | 7                | 21 | 1.3222 | 9.2554  |



|             |                   |    |        |                           |
|-------------|-------------------|----|--------|---------------------------|
| 22.5 – 25.5 | 4                 | 24 | 1.3802 | 5.5208                    |
| 25.5 – 28.5 | 1                 | 27 | 1.4314 | 1.4314                    |
|             | $N = \sum f = 68$ |    |        | $\sum f \log m = 81.9092$ |

$$\text{G.M.} = \text{antilog of } \frac{\sum F \log m}{N}$$

$$\text{G.M.} = \text{antilog } \frac{81.9092}{68} = \text{antilog of } 1.2045$$

$$= 16.02 \text{ maunds}$$

#### Q6 (Continuous Series)

| Class Intervals | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
|-----------------|------|-------|-------|-------|-------|
| Frequencies     | 5    | 8     | 12    | 10    | 5     |

**Solution:**

$$N = 5 + 8 + 12 + 10 + 5 = 40$$

$$\left[ Q_1 = \frac{N}{4} = 10 \right]$$

$$[ Q_3 = \frac{3N}{4} = 30 ]$$

| CLASS INTERVAL | FREQUENCIES | C.F |
|----------------|-------------|-----|
| 0-10           | 5           | 5   |
| 10-20          | 8           | 13  |
| 20-30          | 12          | 25  |
| 30-40          | 10          | 35  |
| 40-50          | 5           | 40  |

$$(Q_1)_{\text{class}} = 10 - 20$$



$$\left[ Q_1 = L + \frac{\left(\frac{N}{4} - cf\right)}{f} \times h \right]$$

$$\left[ Q_1 = 10 + \frac{(10 - 5)}{8} \times 10 \right]$$

$$[Q_1 = 16.25]$$

$$(Q_3)_{class} = 30 - 40$$

$$\left[ Q_3 = 30 + \frac{(30 - 25)}{10} \times 10 \right]$$

$$[Q_3 = 35]$$
$$[IQR = 35 - 16.25 = 18.75]$$
$$\mathbf{IQR = 18.75}$$

**Q7. Compute the standard deviation from the following data:**

| Class     | 0 - 10 | 10 - 20 | 20 - 30 | 30 - 40 | 40-50 |
|-----------|--------|---------|---------|---------|-------|
| Frequency | 5      | 8       | 15      | 16      | 6     |

**SOLUTION:**

| x       | M  | F  | $d = \frac{m-25}{10}$ | d <sup>2</sup> | fd  | fd <sup>2</sup> |
|---------|----|----|-----------------------|----------------|-----|-----------------|
| 0 - 10  | 5  | 5  | -2                    | 4              | -10 | 20              |
| 10 - 20 | 15 | 8  | -1                    | 1              | -8  | 8               |
| 20 - 30 | 25 | 15 | 0                     | 0              | 0   | 0               |



|         |    |                                     |   |   |                                  |                                    |
|---------|----|-------------------------------------|---|---|----------------------------------|------------------------------------|
| 30 - 40 | 35 | 16                                  | 1 | 1 | 16                               | 16                                 |
| 40 - 50 | 45 | 6                                   | 2 | 4 | 12                               | 24                                 |
|         |    | <b><math>N = \sum f = 50</math></b> |   |   | <b><math>\sum fd = 10</math></b> | <b><math>\sum fd^2 = 68</math></b> |

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2 * C}$$

$$\sigma = \sqrt{\frac{68}{50} - \left(\frac{10}{50}\right)^2 * 10}$$

$$= \sqrt{1.36 - (0.2)^2 \times 10}$$

$$= \sqrt{1.36 - 0.04 \times 10}$$

$$= \sqrt{1.32 \times 10}$$

$$= 1.1489 \times 10$$

$$= \mathbf{11.49}$$

Q8. Calculate the **standard deviation AND variance** from the following data;

|                        |    |    |    |    |    |    |
|------------------------|----|----|----|----|----|----|
| <b>Marks</b>           | 10 | 20 | 30 | 40 | 50 | 60 |
| <b>No. of students</b> | 8  | 12 | 20 | 10 | 7  | 3  |

**Solution:** Calculation of standard deviation (from actual mean)

| <b>x</b> | <b>F</b> | <b>Fx</b> | <b><math>d = x - \bar{X} = x - 30.8</math></b> | <b><math>d^2</math></b> | <b><math>fd^2</math></b> |
|----------|----------|-----------|------------------------------------------------|-------------------------|--------------------------|
| 10       | 8        | 80        | -20.8                                          | 432.64                  | 3461.12                  |



|    |                                     |                                    |       |        |                                          |
|----|-------------------------------------|------------------------------------|-------|--------|------------------------------------------|
| 20 | 12                                  | 240                                | -10.8 | 116.64 | 1399.68                                  |
| 30 | 20                                  | 600                                | -0.8  | 0.64   | 12.80                                    |
| 40 | 10                                  | 400                                | 9.2   | 84.64  | 846.40                                   |
| 50 | 7                                   | 350                                | 19.2  | 368.64 | 2580.48                                  |
| 60 | 3                                   | 180                                | 29.2  | 852.64 | 2557.92                                  |
|    | <b>N = <math>\sum f</math> = 60</b> | <b><math>\sum fx</math> = 1850</b> |       |        | <b><math>\sum fd^2</math> = 10858.40</b> |

$$\text{Mean: } \bar{X} = \frac{\sum fx}{N}$$

$$= \frac{1850}{60}$$

$$= 30.8$$

$$\text{Standard Deviation: } \sigma = \sqrt{\frac{\sum fd^2}{N}} = \sqrt{\frac{10858.40}{60}} = 13.45$$